

2.3.17

EE25BTECH11014 - Bhoomika Lokesh

Question: Find the properties of the triangle with vertices $(-4, 0)$, $(0, 3)$ and $(4, 0)$.

Solution:

Name	Point
A	$\begin{pmatrix} -4 \\ 0 \end{pmatrix}$
B	$\begin{pmatrix} 0 \\ 3 \end{pmatrix}$
C	$\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

TABLE 0: Variables Used

We can calculate the magnitude of sides of the triangle using norms:

$$(\mathbf{A} - \mathbf{B}) = \begin{pmatrix} -4 \\ 0 \end{pmatrix} - \begin{pmatrix} 0 \\ 3 \end{pmatrix} = \begin{pmatrix} -4 \\ -3 \end{pmatrix} \quad (0.1)$$

$$|\mathbf{A} - \mathbf{B}| = \sqrt{(\mathbf{A} - \mathbf{B})^T \cdot (\mathbf{A} - \mathbf{B})} \quad (0.2)$$

$$(0.3)$$

$$(\mathbf{A} - \mathbf{B})^T \cdot (\mathbf{A} - \mathbf{B}) = \begin{pmatrix} -4 & -3 \end{pmatrix} \begin{pmatrix} -4 \\ -3 \end{pmatrix} \quad (0.4)$$

$$|\mathbf{A} - \mathbf{B}| = \sqrt{(-4)^2 + (-3)^2} \quad (0.5)$$

$$|\mathbf{A} - \mathbf{B}| = 5 \quad (0.6)$$

Similarly,

$$\mathbf{B} - \mathbf{C} = \begin{pmatrix} 0 \\ 3 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ 3 \end{pmatrix} \quad (0.7)$$

$$|\mathbf{B} - \mathbf{C}| = \sqrt{(\mathbf{B} - \mathbf{C})^T \cdot (\mathbf{B} - \mathbf{C})} \quad (0.8)$$

$$|\mathbf{B} - \mathbf{C}| = \sqrt{(-4)^2 + (3)^2} \quad (0.9)$$

$$|\mathbf{B} - \mathbf{C}| = 5 \quad (0.10)$$

And

$$(\mathbf{A} - \mathbf{C}) = \begin{pmatrix} -4 \\ 0 \end{pmatrix} - \begin{pmatrix} 4 \\ 0 \end{pmatrix} = \begin{pmatrix} -8 \\ 0 \end{pmatrix} \quad (0.11)$$

$$|\mathbf{A} - \mathbf{C}| = \sqrt{(\mathbf{A} - \mathbf{C})^T \cdot (\mathbf{A} - \mathbf{C})} \quad (0.12)$$

$$|\mathbf{A} - \mathbf{C}| = \sqrt{(-8)^2 + (0)^2} \quad (0.13)$$

$$|\mathbf{A} - \mathbf{C}| = 8 \quad (0.14)$$

$$\text{As } |\mathbf{A} - \mathbf{B}|^2 + |\mathbf{B} - \mathbf{C}|^2 = 25 + 25 = 50 \quad (0.15)$$

$$\text{and } |\mathbf{A} - \mathbf{C}|^2 = 64 \quad (0.16)$$

Hence the triangle is **not RIGHT** angled.

Angle at Vertex A ($\angle A$)

$$(\mathbf{A} - \mathbf{B})^T \cdot (\mathbf{A} - \mathbf{C}) = \begin{pmatrix} -4 & 3 \end{pmatrix} \begin{pmatrix} -8 \\ 0 \end{pmatrix} = 32$$

$$\cos(\angle A) = \frac{(\mathbf{A} - \mathbf{B})^T \cdot (\mathbf{A} - \mathbf{C})}{|\mathbf{A} - \mathbf{B}| |\mathbf{A} - \mathbf{C}|} \quad (0.17)$$

$$\cos(\angle A) = \frac{32}{5 \cdot 8} = 0.8 \quad (0.18)$$

$$\Rightarrow \angle A = \cos^{-1}(0.8) \approx 36.87^\circ \quad (0.19)$$

Angle at Vertex B ($\angle B$)

$$(\mathbf{B} - \mathbf{A})^T \cdot (\mathbf{B} - \mathbf{C}) = \begin{pmatrix} -4 & -3 \end{pmatrix} \begin{pmatrix} 4 \\ -3 \end{pmatrix} = -16 + 9 = -7 \quad (0.20)$$

$$\cos(\angle B) = \frac{-7}{5 \cdot 5} = -0.28 \quad (0.21)$$

$$\Rightarrow \angle B = \cos^{-1}(-0.28) \approx 106.26^\circ \quad (0.22)$$

Angle at Vertex C ($\angle C$)

$$(\mathbf{C} - \mathbf{A})^T \cdot (\mathbf{C} - \mathbf{B}) = \begin{pmatrix} -8 & 0 \end{pmatrix} \begin{pmatrix} -4 \\ 3 \end{pmatrix} = 32 \quad (0.23)$$

$$\cos(\angle C) = \frac{32}{8 \cdot 5} = 0.8 \quad (0.24)$$

$$\Rightarrow \angle C = \cos^{-1}(0.8) \approx 36.87^\circ \quad (0.25)$$

$$\therefore \angle A \approx 36.87^\circ, \quad \angle B \approx 106.26^\circ, \quad \angle C \approx 36.87^\circ$$

$\therefore$ We can conclude that triangle **ABC** is an **ISOSCELES** triangle.

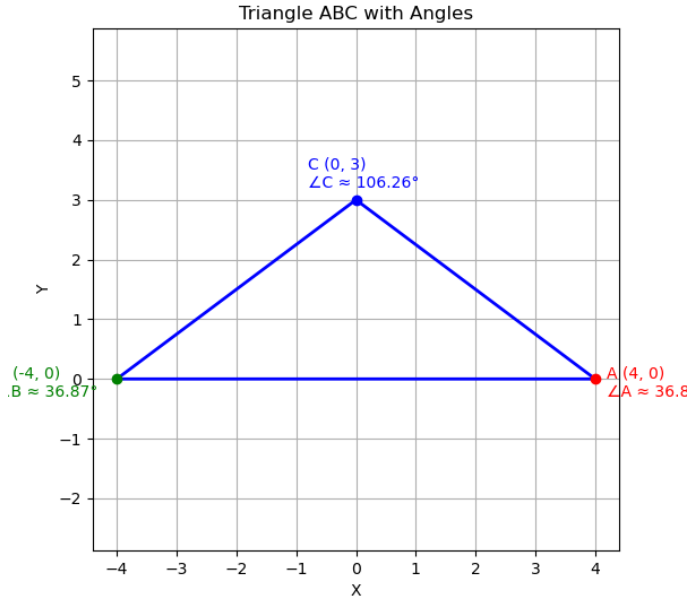


Fig. 0.1: Graph